

Instructor: Dr. Rajesh Kumble Nayak
 10:00 AM, 02 November, 2024.
 Duration 150 Min.

- Read each question carefully.
- Answer all questions.
- Ensure all parts of the answer for a given question are in the same place.
- Write neatly and do all rough work on the answer sheet.
- Do not write on the question paper.
- Keep answers concise and to the point.

Good luck!

Q - 1(20 Marks)

For two functions $f(q, p, t)$ and $g(q, p, t)$, where (q, p) are canonical variable and t is time, show that,

$$\frac{d}{dt} [f, g] = \left[\frac{df}{dt}, g \right] + \left[f, \frac{dg}{dt} \right],$$

Here, $[f, g]$ is a Poisson bracket *w.r.t* variables (q, p) .

Q - 2(20 Marks)

For the Hamiltonian,

$$H = q_1 p_1 - q_2 p_2 - a q_1^2 + b q_2^2,$$

with a and b are constants,

1. Show that for a time-independent function of canonical variables, $G(p, q)$, if G is constant of motion, then

$$[G, H] = 0,$$

where $H(p, q)$ is Hamiltonian.

2. Show that,

$$F_1 = \frac{p_1 - a q_1}{q_2}, \quad \text{and} \quad F_2 = q_1 q_2,$$

are constants of motion.

3. Can you find more conserved quantities using Jacobies-identity?

Jacobies-identity look's like, $:= [X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$, where, $[X, Y]$ is a Poisson bracket *w.r.t* variables (q, p) .

Q - 3(20 Marks)

Consider the Kepler problem, i.e. $V(r) = -\frac{k}{r}$.

1. Find the effective radial equation of motion.

P.T.O

2. For a bound orbit, the distance of closest approach (r_{min}) and farthest passage (r_{max}) of the planet is determined by the condition $\dot{r} = 0$. Find r_{min} and r_{max} in terms of E and L_z .
3. Mean distance is the length of semi-major axis, $a = (r_{min} + r_{max})/2$ and show that it depends only on the Energy of the particle!
4. For an artificial satellite, how conserved quantities such as E and L_z can be used to change the orbital parameters?

Q - 4(20 Marks)

For the Hamiltonian given below:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2$$

1. Find an explicit canonical transformation $(q, p) \rightarrow (Q, P)$ such that Q is cyclic
2. Find the new Hamiltonian K and conserved momenta
3. Express the solution for EOM in canonical coordinates (q, p)

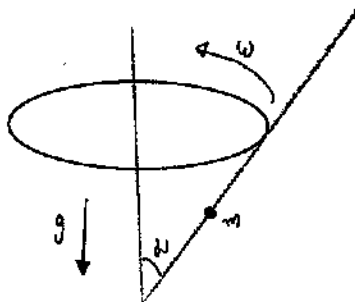
Q - 5(20 Marks)

The angular momentum is given by $\vec{L} = \vec{r} \times \vec{p}$ compute the component of angular momentum in Cartesian coordinate system $\vec{L} = (L_x, L_y, L_z)$, where $\vec{p} = m\vec{v}$, in terms of suitable canonical variables (q, p) .

1. Compute the Poisson bracket $[L_x, L_y]$, $[L_y, L_z]$ and $[L_z, L_x]$ for the suitable canonical variables.

Q - 6(20 Marks)

A bead of mass m slides on a long straight wire, making an angle α with, and rotates with constant angular velocity ω about, the upward vertical. As shown in the figure below. Gravity g acts vertically downwards!



1. Choose an appropriate generalised coordinate and find the Lagrangian.
2. Write down Euler-Lagrange's equation of motion as the second-order ODE in generalised coordinates
3. Obtain the equation of motion from the energy conservation.
4. Are the equations of motion obtained from (1) and (2) are same? If not, why?

- The End -

Rayan